Experiment

I^d PhO 2020

A1-1

English (Official)

A. From slit to crystal

A.1 (0.3 pt)

$I(q) =$

A.2 (0.2 pt)

$q =$

A.3 (0.2 pt)

$q(q_1) =$

Expression for q_1 and a:

A.4 (1.0 pt)
Formulas, schemes and pictures for explanation:

DG1: $q_1 =$ $a =$

DG2: $q_1 =$ $a =$

DG3: $q_1 =$ $a =$

DG4: $q_1 =$ $a =$

DG5: $q_1 =$ $a =$

A.5 (1.5 pt)
Formulas, schemes and pictures for explanation:

DG3: $a/b =$

DG4: $a/b =$

DG5: $a/b =$

A.6 (0.7 pt)

$\rho(x) =$

$F_A(h) =$

$h =$

A.7 (0.7 pt)

$F_B(h) =$

$h =$

A.8 (0.4 pt)

$\frac{I_{A,h=0}}{I_{B,h=0}} =$

$\frac{I_{A,h=1}}{I_{B,h=1}} =$

B. 2D crystal

B.1 (1 pt)

$q_1 =$

$q_2 =$

$\beta =$

B.2 (1.0 pt)

Crystal A $|F(h,k)| =$

Crystal D $|F(h,k)| =$

B.3 (0.6 pt)

$a_{UC1} =$

$a_{UC2} =$

$a_{UC3} =$

$a_{UC4} =$

B.4 (0.4 pt)

UC1 – UC2 – UC3 – UC4 –

B.5 (0.8 pt)

$b =$

B.6 (1.2 pt)

UC5

$a_1 =$

$a_2 =$

$\alpha =$

UC6

$a_1 =$

$a_2 =$

$\alpha =$

UC7

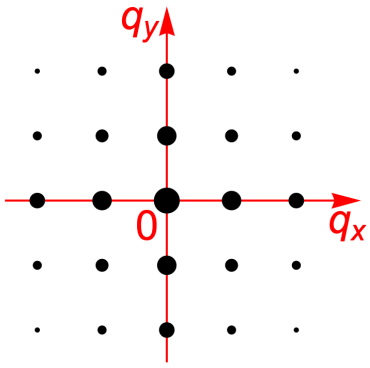
$a_1 =$

$a_2 =$

$\alpha =$

C. Symmetries of crystals

C.1 (0.3 pt)



$h =$ $k =$

$m =$

C.2 (0.2 pt)

Equations for all possible axis symmetry:

C.3 (0.4 pt)
designation (C_m for rotational and equations for axis symmetry) and equation on intensities $I(q_x, q_y)$

C $I(q_x, q_y) =$

C $I(q_x, q_y) =$

C $I(q_x, q_y) =$

$I(q_x, q_y) =$

$I(q_x, q_y) =$

$I(q_x, q_y) =$

$I(q_x, q_y) =$

C.4 (0.2 pt)

$I(-h, -k) =$

Which symmetry corresponds?

C.5 (0.4 pt)

$f_2(q_x, q_y) =$

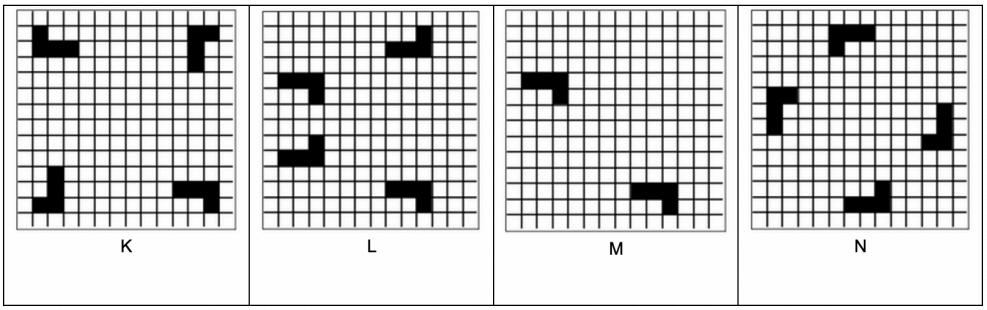
$f_3(q_x, q_y) =$

$f_4(q_x, q_y) =$

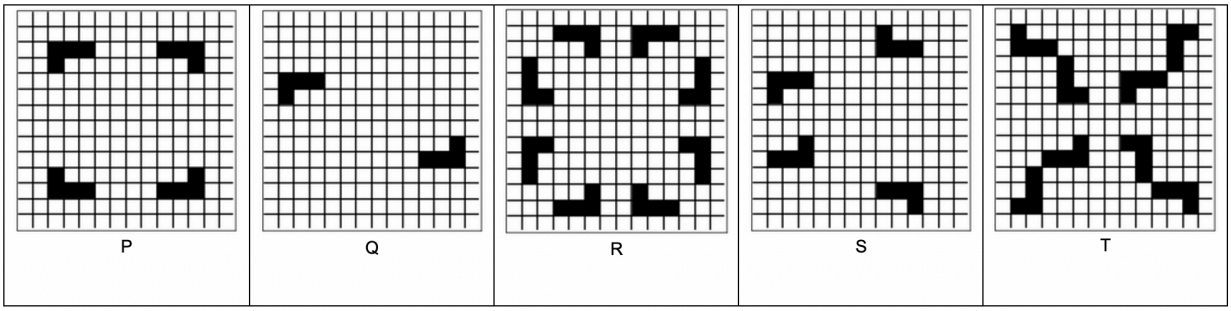
C.6 (0.5 pt)
List all possible rotational symmetries:

$m =$

C.7 (0.9 pt)



C.7 (cont.)



C.8 (0.8 pt)

Use this table if you need to note any information about samples.

PG									
1									
2									
5									
8									

PG1 – PG2 – PG5 – PG8 –

C.9 (1.0 pt)
Use this table if you need to note any information about samples.

PG									
3									
4									
6									
7									
9									

PG3 – PG4 – PG6 – PG7 – PG9 –

C.10 (0.3 pt)

Could it be crystal? Circle your option: Yes / No

D. You'll need the phases. . .

D.1 (1 pt)

$I_{MR0} =$

$I_{MR2} =$

D.2 (2.0 pt)

MR1:

D.3 (2.0 pt)
MR2: